Data Dictionary

- `Zone.csv`
- `Whole_Bldg.csv`
- `Envelope.csv`

`Zone.csv`

General Information

#	Database Column Name	Value Type	Description
1	**Prototype**	*Text/String*	Building prototype name (e.g., Assembly, RetailMedium). Use to filter which rows apply to which model/run.
2	**Vintage**	*Text/String*	Code/era bucket (NC, V4, V3-2-1). Use to choose which vintage apply.

Zone Info (Object Class_1)

#	Database Column Name	Value Type	Description
3	**Object Class_1**	*Text/String*	The EnergyPlus Class type: **ZONE**
4	**Object Name_1.1**	*Text/String*	The EnergyPlus object name for that object class (e.g., Zone name).
5	**#SpaceType**	*Text/String*	Space-type category label (e.g., "Convention, Conference..."). Use mainly for grouping and naming.
6	**Field Name**	*Text/String*	Multiplier: A field in the Zone class.
7	**Value**	*Numeric*	Multiplier value for the specific Zone in the prototype model
8	**Unit**	*Text/String*	n/a (Multiplier doesn't have a unit)
9	**Field Name**	*Text/String*	Floor_Area: A field in the Zone class.
10	**Value**	*Numeric*	Zone square footage value in the prototype model
11	**Unit**	*Text/String*	Square Foot
12	**Field Name**	*Text/String*	Part_of_Total_Floor_Area: A field in the Zone class. Use to calculate the total conditioned floor area in the prototype model.
13	**Value**	*Text/String*	Yes/No

People Info (Object Class_2)

#	Database Column Name	Value Type	Description
14	**Object Class_2**	*Text/String*	The EnergyPlus class type: **PEOPLE**
15	**Object Name_2.1**	*Text/String*	Name of the People object for this zone/space (e.g., People_Exhibits and events_Conference Room). Used so the zone can reference this internal load.
16	**Number_of_People_Schedule_Name**	*Text/String*	Schedule that controls when people are present (fraction 0–1) in the zone during the day.
17	**Number_of_People_Calculation_Method**	*Text/String*	How the people count is defined (e.g., People/Area).
18	**People_per_Floor_Area**	*Numeric*	People density input when method is People/Area (units depend on model convention; typically people/ft^2).
19	**Activity_Level_Schedule_Name**	*Text/String*	Schedule for occupant activity (metabolic rate). Maps to People, Activity Level Schedule Name.

Light Info (Object Class_3)

#	Database Column Name	Value Type	Description
20	**Object Class_3**	*Text/String*	The EnergyPlus Class type: **LIGHT**
21	**Object Name_3.1**	*Text/String*	Name of the Lights object (e.g., Lights_Theater Area...). Defines lighting internal gains for the zone/space.
22	**Schedule_Name**	*Text/String*	Lighting schedule (fraction 0–1) controlling the lighting load in the zone. Maps to Lights, Schedule Name.
23	**Design_Level_Calculation_Method**	*Text/String*	Method used to define lighting power (e.g., Watts/Area). Maps to Lights, Design Level Calculation Method.
24	**Lights_Wattes_per_Floor_Area**	*Numeric*	Lighting power density value when method is Watts/Area. Maps to Lights, Watts per Zone Floor Area.
25	**Unit**	*Text/String*	Units label for the lighting input (commonly W/ft^2 in zone.csv; used for conversion/QA).
26	**End_Use_Subcategory_3.1**	*Text/String*	End-use tag for reporting (e.g., ComplianceLtg). Maps to Lights, End-Use Subcategory.

Electric Equipment Info (Object Class_4)

#	Database Column Name	Value Type	Description
27	**Object Class_4**	*Text/String*	The EnergyPlus Class type: **ELECTRICEQUIPMENT**
28	**Object Name_4.1**	*Text/String*	Name of the plug-load ElectricEquipment object (receptacle/plug load). Used to define general electric internal gains.
29	**Schedule_Name**	*Text/String*	Plug-load schedule (fraction 0–1) in the zone. Maps to ElectricEquipment, Schedule Name.
30	**Design_Level_Calculation_Method**	*Text/String*	Method used to define plug load (e.g., Watts/Area). Maps to ElectricEquipment, Design Level Calculation Method.
31	**Equip_Wattes_per_Floor_Area**	*Numeric*	Plug-load power density value when method is Watts/Area. Maps to ElectricEquipment, Watts per Zone Floor Area.
32	**Unit**	*Text/String*	Units label for plug-load input (commonly W/ft^2).
33	**End_Use_Subcategory_4.1**	*Text/String*	End-use tag for reporting (e.g., Receptacle). Maps to ElectricEquipment, End-Use Subcategory.
34	**Object Name_4.2**	*Text/String*	Name of the refrigeration load object (often modeled as a second ElectricEquipment object).
35	**Schedule_Name**	*Text/String*	Refrigeration schedule (fraction 0–1) in the zone. Maps to ElectricEquipment, Schedule Name for that refrigeration object.
36	**Design_Level_Calculation_Method**	*Text/String*	Method used (e.g., Watts/Area). Maps to ElectricEquipment, Design Level Calculation Method.
37	**Refrig_Wattes_per_Floor_Area**	*Numeric*	Refrigeration power density value when method is Watts/Area. Maps to ElectricEquipment, Watts per Zone Floor Area.
38	**Unit**	*Text/String*	Units label (commonly W/ft^2 in zone.csv; used for conversion/QA).
39	**End_Use_Subcategory_4.2**	*Text/String*	End-use tag for reporting (e.g., Refrigeration). Maps to ElectricEquipment, End-Use Subcategory.

Gas Equipment Info (Object Class_5)

#	Database Column Name	Value Type	Description
40	**Object Class_5**	*Text/String*	The EnergyPlus Class type: **GASEQUIPMENT**

#	Database Column Name	Value Type	Description
41	**Object Name_5.1**	*Text/String*	Name of the GasEquipment object (e.g., cooking/process gas loads if any).
42	**Schedule_Name**	*Text/String*	Gas equipment schedule (fraction 0–1) in the zone. Maps to GasEquipment, Schedule Name.
43	**Design_Level_Calculation_Method**	*Text/String*	Method used to define the gas load (e.g., Power/Area). Maps to GasEquipment, Design Level Calculation Method.
44	**Gas_Power_per_Floor_Area**	*Numeric*	Gas load density when method is Power/Area. Maps to GasEquipment, Power per Zone Floor Area.
45	**Unit**	*Text/String*	Units label for gas power density (often Btu/hr-ft^2 in Zone.CSV).
46	**End_Use_Subcategory_5.1**	*Text/String*	End-use tag for reporting (e.g., Gas). Maps to GasEquipment, End-Use Subcategory.

Zone Infiltration Info (Object Class_6)

#	Database Column Name	Value Type	Description
47	**Object Class_6**	*Text/String*	The EnergyPlus Class type: **ZONEINFILTRATION:DESIGNFLOWRATE**
48	**Object Name_6.1**	*Text/String*	Name of the infiltration object for the zone.
49	**Schedule_Name**	*Text/String*	Infiltration schedule (fraction 0–1) in the zone. Maps to ZoneInfiltration:DesignFlowRate, Schedule Name.
50	**Design_Level_Calculation_Method**	*Text/String*	Infiltration method (e.g., Flow/ExteriorWallArea or AirChanges/Hour). Maps to ZoneInfiltration:DesignFlowRate, Design Flow Rate Calculation Method.
51	**Infil_Air_Change_per_hour**	*Numeric*	ACH value used only when method is AirChanges/Hour. Maps to ZoneInfiltration:DesignFlowRate, Air Changes per Hour.

#	Database Column Name	Value Type	Description
52	**Infil_Flow_Rate_per_Exterior_Surface_Area**	*Numeric*	Flow per exterior surface area used when method is Flow/ExteriorWallArea. Maps to ZoneInfiltration:DesignFlowRate, Flow per Exterior Surface Area.

Zone Ventilation Info (Object Class_7)

#	Database Column Name	Value Type	Description
53	**Object Class_7**	*Text/String*	The EnergyPlus Class type: **DESIGNSPECIFICATION:OUTDOORAIR**
54	**Object Name_7.1**	*Text/String*	Name of the outdoor air specification object tied to the zone.
55	**Outdoor_Air_Method**	*Text/String*	How OA is calculated (e.g., Maximum). Maps to DesignSpecification:OutdoorAir, Outdoor Air Method.
56	**Outdoor_Airflow_per_Person**	*Numeric*	OA rate per personin the zone. Maps to DesignSpecification:OutdoorAir, Outdoor Air Flow per Person.
57	**Outdoor_Airflow_per_Zone_Floor_Area**	*Numeric*	OA rate per floor area in the zone. Maps to DesignSpecification:OutdoorAir, Outdoor Air Flow per Zone Floor Area.
58	**Outdoor_Airflow_per_Zone**	*Numeric*	Fixed OA flow per zone. Maps to DesignSpecification:OutdoorAir, Outdoor Air Flow per Zone.
59	**Outdoor_Air_Schedule_Name**	*Text/String*	Availability schedule that can scale OA (fraction 0–1). Maps to DesignSpecification:OutdoorAir, Outdoor Air Schedule Name.

Water Use Equipment Info (Object Class_8)

#	Database Column Name	Value Type	Description
60	**Object Class_8**	*Text/String*	The EnergyPlus Class type: **WATERUSE:EQUIPMENT**
61	**Object Name_8.1**	*Text/String*	Name of the zone/space hot-water end-use object.

#	Database Column Name	Value Type	Description
62	**#Water_Heater_Type**	*Text/String*	Water heater classification (Gas/Electric/HP). Used by the workflow to choose which water heater + plant loop objects to generate; not a direct WaterUse:Equipment numeric field.
63	**Flow_Rate_Fraction_Schedule_Name**	*Text/String*	Schedule (fraction 0–1) scaling hot water draw. Maps to WaterUse:Equipment, Flow Rate Fraction Schedule Name.
64	**Target_Temperature_Schedule_Name**	*Text/String*	Schedule defining desired hot water temp. Maps to WaterUse:Equipment, Target Temperature Schedule Name.
65	**Peak_Flow_Rate**	*Numeric*	Peak hot water flow rate for this end use. Maps to WaterUse:Equipment, Peak Flow Rate.

Thermostat Setpoint Info (Object Class_9)

#	Database Column Name	Value Type	Description
66	**Object Class_9**	*Text/String*	The EnergyPlus Class type: **THERMOSTATSETPOINT:DUALSETPOINT**
67	**Object Name_9.1**	*Text/String*	Name of the dual setpoint thermostat object assigned to the zone.
68	**Heating_SetPoint_Schedule_Name**	*Text/String*	Heating setpoint schedule. Maps to ThermostatSetpoint:DualSetpoint, Heating Setpoint Temperature Schedule Name.
69	**Cooling_SetPoint_Schedule_Name**	*Text/String*	Cooling setpoint schedule. Maps to ThermostatSetpoint:DualSetpoint, Cooling Setpoint Temperature Schedule Name.

`Whole_Bldg.csv`

General Information

#	Database Column Name	Value Type	Description
1	**Prototype**	*Text/String*	Building prototype name (e.g., Assembly, RetailMedium). Use to filter which rows apply to which model/run.
2	**Vintage**	*Text/String*	Code/era bucket (NC, V4, V3-2-1). Use to choose which vintage apply.

Ext. Lights Info (Object Class_1)

#	Database Column Name	Value Type	Description
3	**Object Class_1**	*Text/String*	The EnergyPlus Class type: **EXTERIOR:LIGHTS**
4	**Object Name_1.1**	*Text/String*	The EnergyPlus object Name for that object class (e.g., exterior lighting object name).
5	**Schedule_Name**	*Text/String*	The schedule that controls when lights are on; typically 0–1. Maps to Ext:Lights, Schedule Name.
6	**Design_Level**	*Text/String*	Method used to define Ext. lighting power, full power when schedule = 1.(e.g., Watts/Area). Maps to Exterior:Lights, Design Level.
7	**Unit**	*Numeric*	Units for Design_Level (typically W).
8	**Control_Option**	*Text/String*	Ext Light Control Option (e.g., AstronomicalClock, ScheduleNameOnly).
9	**End_Use_Subcategory_1.1**	*Text/String*	End-use tag for reporting : Exterior:Lights, End-Use Subcategory.

Ext. Fuel Equipment Info (Object Class_2)

#	Database Column Name	Value Type	Description
10	**Object Class_2**	*Text/String*	The EnergyPlus Class type: **EXTERIOR:FUELEQUIPMENT**
11	**Object Name_2.1**	*Text/String*	The EnergyPlus object Name for that object class (e.g., elevator load object name).
12	**Fuel_Use_Type**	*Text/String*	Fuel Use Type (e.g., Electricity, NaturalGas).
13	**Schedule_Name**	*Text/String*	The schedule that scales the load over time; typically 0–1, Maps to Exterior:FuelEquipment, Schedule Name ().

#	Database Column Name	Value Type	Description
14	**Design_Level**	*Numeric*	Method used to define Ext.FuelEquipment, Design Level = peak rate when schedule = 1.
15	**Unit**	*Text/String*	Units for Design_Level (typically W for Electricity).
16	**End_Use_Subcategory_2.1**	*Text/String*	End-use tag for reporting: Exterior:FuelEquipment, End-Use Subcategory (reporting label, e.g., Elevators).

Water Heater Info (Object Class_3)

#	Database Column Name	Value Type	Description
17	**Object Class_3**	*Text/String*	The EnergyPlus Class type: **WATERHEATER:MIXED**
18	**Object Name_3.1**	*Text/String*	The EnergyPlus object Name for that object class (e.g., NonResBaseGasWaterHeater).
19	**Tank_Volume**	*Numeric*	Tank Volume.
20	**Unit**	*Text/String*	Units for tank volume (e.g., gal).
21	**Setpoint_temperature_Schedule_Name**	*Text/String*	The schedule for the water heater setpoint temperature .
22	**Heater_Maximum_Capacity**	*Numeric*	Heater Maximum Capacity.
23	**Unit**	*Text/String*	Units for heater maximum capacity (typically Btu/hr).
24	**Ambient_temperature_Indicator**	*Text/String*	Ambient Temperature Indicator (e.g., Schedule, Zone, Outdoors).
25	**Ambient_temperature_Schedule_Name**	*Text/String*	The schedule for the Ambient Temperature (used when Ambient Temperature Indicator = Schedule).
26	**Off_On_Cycle_Loss_Coefficient_to_Ambient_Temperature**	*Numeric*	Off/On Cycle Loss Coefficient to Ambient Temperature (standby loss coefficient to ambient).
27	**Unit**	*Text/String*	Units for loss coefficient (Typically Btu/hr-F).

#	Database Column Name	Value Type	Description
28	**End_Use_Subcategory_3.1**	*Text/String*	End-use tag for reporting (commonly “GasWaterHeater”).

Envelope.csv

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1	Prototype	CZ	Vintage	Object Class_1	Object Name_1.1 / FixedWindow_Nonres	Field Name	Value	Unit	Field Name	Value	Unit	Field Name	Value	Unit
2	Assembly	1	NC	WINDOWMATERIAL:SIMPLEGLAZINGSYSTEM	FixedWindow_Nonres	Ufactor	0.36	Btu/h-ft2-F	Solar_Heat_Gain_Coefficient	0.25	n/a	Visible_Transmittance	0.42	n/a
3	Assembly	2	NC	WINDOWMATERIAL:SIMPLEGLAZINGSYSTEM	FixedWindow_Nonres	Ufactor	0.36	Btu/h-ft2-F	Solar_Heat_Gain_Coefficient	0.25	n/a	Visible_Transmittance	0.42	n/a
4	Assembly	3	NC	WINDOWMATERIAL:SIMPLEGLAZINGSYSTEM	FixedWindow_Nonres	Ufactor	0.36	Btu/h-ft2-F	Solar_Heat_Gain_Coefficient	0.25	n/a	Visible_Transmittance	0.42	n/a
5	Assembly	4	NC	WINDOWMATERIAL:SIMPLEGLAZINGSYSTEM	FixedWindow_Nonres	Ufactor	0.36	Btu/h-ft2-F	Solar_Heat_Gain_Coefficient	0.25	n/a	Visible_Transmittance	0.42	n/a
6	Assembly	5	NC	WINDOWMATERIAL:SIMPLEGLAZINGSYSTEM	FixedWindow_Nonres	Ufactor	0.36	Btu/h-ft2-F	Solar_Heat_Gain_Coefficient	0.25	n/a	Visible_Transmittance	0.42	n/a
7	Assembly	6	NC	WINDOWMATERIAL:SIMPLEGLAZINGSYSTEM	FixedWindow_Nonres	Ufactor	0.36	Btu/h-ft2-F	Solar_Heat_Gain_Coefficient	0.25	n/a	Visible_Transmittance	0.42	n/a
8	Assembly	7	NC	WINDOWMATERIAL:SIMPLEGLAZINGSYSTEM	FixedWindow_Nonres	Ufactor	0.36	Btu/h-ft2-F	Solar_Heat_Gain_Coefficient	0.25	n/a	Visible_Transmittance	0.42	n/a
9	Assembly	8	NC	WINDOWMATERIAL:SIMPLEGLAZINGSYSTEM	FixedWindow_Nonres	Ufactor	0.36	Btu/h-ft2-F	Solar_Heat_Gain_Coefficient	0.25	n/a	Visible_Transmittance	0.42	n/a
10	Assembly	9	NC	WINDOWMATERIAL:SIMPLEGLAZINGSYSTEM	FixedWindow_Nonres	Ufactor	0.34	Btu/h-ft2-F	Solar_Heat_Gain_Coefficient	0.22	n/a	Visible_Transmittance	0.42	n/a
11	Assembly	10	NC	WINDOWMATERIAL:SIMPLEGLAZINGSYSTEM	FixedWindow_Nonres	Ufactor	0.36	Btu/h-ft2-F	Solar_Heat_Gain_Coefficient	0.25	n/a	Visible_Transmittance	0.42	n/a
12	Assembly	11	NC	WINDOWMATERIAL:SIMPLEGLAZINGSYSTEM	FixedWindow_Nonres	Ufactor	0.34	Btu/h-ft2-F	Solar_Heat_Gain_Coefficient	0.22	n/a	Visible_Transmittance	0.42	n/a
13	Assembly	12	NC	WINDOWMATERIAL:SIMPLEGLAZINGSYSTEM	FixedWindow_Nonres	Ufactor	0.34	Btu/h-ft2-F	Solar_Heat_Gain_Coefficient	0.22	n/a	Visible_Transmittance	0.42	n/a
14	Assembly	13	NC	WINDOWMATERIAL:SIMPLEGLAZINGSYSTEM	FixedWindow_Nonres	Ufactor	0.34	Btu/h-ft2-F	Solar_Heat_Gain_Coefficient	0.22	n/a	Visible_Transmittance	0.42	n/a

BC	BD	BE	BF	BG	BH	BI
15	16	17	18	19	20	21
Object Class_2	Object Name_2.1 / Insulation_ExtWall_MetalFramed_Nonres	Field Name	Value	Unit	#U_Factor	Unit
MATERIAL	Insulation_ExtWall_MetalFramed_Nonres	Thickness	2.29225	in.	0.06	BTU/(h-ft2-F)
MATERIAL	Insulation_ExtWall_MetalFramed_Nonres	Thickness	2.533159091	in.	0.055	BTU/(h-ft2-F)
MATERIAL	Insulation_ExtWall_MetalFramed_Nonres	Thickness	1.88168662	in.	0.071	BTU/(h-ft2-F)
MATERIAL	Insulation_ExtWall_MetalFramed_Nonres	Thickness	2.533159091	in.	0.055	BTU/(h-ft2-F)
MATERIAL	Insulation_ExtWall_MetalFramed_Nonres	Thickness	2.533159091	in.	0.055	BTU/(h-ft2-F)
MATERIAL	Insulation_ExtWall_MetalFramed_Nonres	Thickness	2.29225	in.	0.06	BTU/(h-ft2-F)
MATERIAL	Insulation_ExtWall_MetalFramed_Nonres	Thickness	2.29225	in.	0.06	BTU/(h-ft2-F)
MATERIAL	Insulation_ExtWall_MetalFramed_Nonres	Thickness	2.533159091	in.	0.055	BTU/(h-ft2-F)
MATERIAL	Insulation_ExtWall_MetalFramed_Nonres	Thickness	2.533159091	in.	0.055	BTU/(h-ft2-F)
MATERIAL	Insulation_ExtWall_MetalFramed_Nonres	Thickness	2.533159091	in.	0.055	BTU/(h-ft2-F)
MATERIAL	Insulation_ExtWall_MetalFramed_Nonres	Thickness	2.533159091	in.	0.055	BTU/(h-ft2-F)
MATERIAL	Insulation_ExtWall_MetalFramed_Nonres	Thickness	2.533159091	in.	0.055	BTU/(h-ft2-F)
MATERIAL	Insulation_ExtWall_MetalFramed_Nonres	Thickness	2.533159091	in.	0.055	BTU/(h-ft2-F)

General Information

#	Database Column Name	Value Type	Description
1	**Prototype**	*Text/String*	Building prototype name (e.g., Assembly, RetailMedium). Use to filter which rows apply to which model/run.
2	**CZ**	*Text/String*	California climate zone number used to select the correct envelope values for that location.
3	**Vintage**	*Text/String*	Code/era bucket (NC, V4, V3-2-1). Use to choose which vintage apply.

Glazing Info (Object Class_1)

#	Database Column Name	Value Type	Description

#	Database Column Name	Value Type	Description
4	**Object Class_1**	*Text/String*	The EnergyPlus Class type: **WINDOWMATERIAL:SIMPLEGLAZINGSYSTEM**
5	**Object Name_1.1**	*Text/String*	The EnergyPlus object Name for the glazing system (used by window constructions assigned to nonres windows).
6	**Field Name**	*Text/String*	U-factor: A field in the WindowMaterial:SimpleGlazingSystem.
7	**Value**	*Numeric*	U-factor value for this glazing system.
8	**Unit**	*Text/String*	Units for U-factor (Commonly Btu/h-ft2-F).
9	**Field Name**	*Text/String*	Solar_Heat_Gain_Coefficient:A field in the WindowMaterial:SimpleGlazingSystem.
10	**Value**	*Numeric*	SHGC value.
11	**Unit**	*Text/String*	n/a
12	**Field Name ()**	*Text/String*	Visible_Transmittance: A field in the WindowMaterial:SimpleGlazingSystem.
13	**Value**	*Numeric*	Visible transmittance value
14	**Unit**	*Text/String*	n/a (dimensionless).

> Different glazing types are listed under Object Name_1.2 through Object Name_1.5 using the same format immediately after this section. We show Object Name_1.1 as the example.

Opaque Envelope Info (Object Class_2)

#	Database Column Name	Value Type	Description
15	**Object Class_2**	*Text/String*	The EnergyPlus Class type: **MATERIAL**
16	**Object Name_2.1**	*Text/String*	The EnergyPlus object Name for the insulation layer (used in the nonres metal-framed exterior wall construction).
17	**Field Name**	*Text/String*	Thickness: A field in the Material
18	**Value**	*Numeric*	Thickness of the insulation material used in the corresponding construction assembly.
19	**Unit**	*Text/String*	Units for thickness (in.)
20	**#U_Factor**	*Text/String*	Target/derived U-factor used by our workflow for calibration/QA (often used to back-calculate required insulation thickness or to validate assemblies).
21	**Unit**	*Text/String*	Units for U-factor (Typically BTU/(h-ft2-F)).

> Different wall and roof assembly types are listed under Object Name_2.1 through Object Name_2.14 in the same format immediately after this section. We show Object Name_2.1 as the example.